symmetry between one location and another, that allows us to speak sensibly of time when describing the entire universe.

Subtle Features of an Expanding Universe

A few subtle points in our explanation of cosmic expansion are worthy of emphasis. First, remember that in the balloon metaphor, it is only the balloon's *surface* that plays any role—a surface that is only two-dimensional (each location can be specified by giving two numbers analogous to latitude and longitude on earth), whereas the space we see when we look around has three dimensions. We make use of this lower-dimensional model because it retains the concepts essential to the true, three-dimensional story but is far easier to visualize. It's important to bear this in mind, especially if you have been tempted to say that there *is* a special point in the balloon model: the center point in the interior of the balloon away from which the whole rubber surface is moving. While this observation is true, it is meaningless in the balloon analogy because any point not on the balloon's surface plays no role. The surface of the balloon represents *all* of space; points that do not lie on the surface of the balloon are merely irrelevant by-products of the analogy and do not correspond to any location in the universe.[19]

Second, if the speed of recession is larger and larger for galaxies that are farther and farther away, doesn't that mean that galaxies that are sufficiently distant will rush away from us at a speed greater than the speed of light? The answer is a resounding, definite yes. Yet there is no conflict with special relativity. Why? Well, it's closely related to the reason clocks moving apart due to the cosmic flow of space stay synchronized. As we emphasized in Chapter 3, Einstein showed that nothing can move *through* space faster than light. But galaxies, on average, hardly move through space at all. Their motion is due almost completely to the *stretching of space itself.* And Einstein's theory does not prohibit space from expanding in a way that drives two points—two galaxies—away from each other at greater than light speed. His results only constrain speeds for which motion from spatial expansion has been subtracted out, motion in excess of that arising from spatial expansion. Observations confirm that for typical galaxies zipping along with the